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Abstract

Distribution, abundance estimates, and run timing information was collected on Summer Chum Salmon (*Oncorhynchus keta*) in the Koyukuk River drainage using radio telemetry, spaghetti tags, and the Gisasa River weir during 2015. Tagging efforts used a standardized soak time of 105 minutes per boat per day, five days per week, from 22 June to 30 July by drifting gill nets (10.2 cm x 18.3m x 3m) deployed from boats by two person crews on the lower Koyukuk River. Captured Summer Chum Salmon received a spaghetti tag and every seventh captured Chum Salmon received a radio transmitter. A total of 1,330 fish were spaghetti tagged throughout the summer with 250 fish receiving radio transmitters. Fixed station receivers and aerial telemetry were used to relocate the fish as they entered spawning locations. The Huslia (29%), Koyukuk (24%), Hogatza (9%), Gisasa (8%), Indian (2%), and Kateel (2%), Dulbi (1%) and Alatna (1%) Rivers, along with Henshaw (13%), and Dakli Wheeler (11%) Creeks had fish with transmitters located within their watershed. Recapture of tagged fish occurred at the Gisasa River (64) and Henshaw Creek (74) weirs. The Chapman two-occasion abundance estimate was 855,485 Chum Salmon while the time-stratified Lincoln-Peterson model produced an estimated 969,975.

Introduction

Koyukuk River Summer Chum Salmon (*Oncorhynchus keta*) stocks are one of the largest contributors to the Yukon River Summer Chum Salmon (here after referred to as Chum Salmon) population, yet accurate information on their overall abundance and distribution is severely lacking. These salmon stocks migrate through a number of National Wildlife Refuges (NWR) along the Yukon and Koyukuk Rivers including the Yukon Delta NWR, Koyukuk NWR, Innoko NWR, and Kanuti NWR providing a subsistence resource to people residing in the region. Overall, returns throughout the Yukon River drainage have remained relatively constant; however, there has been a shift in production among rivers. For example, the Anvik River (lower Yukon River) once accounted for 40% of the Chum Salmon returning to the Yukon River, and has declined to less than 25% from 2002 to 2013 (McEwen 2015). Conversely, the Koyukuk River has experienced an increase in escapements in its tributaries over the past 12 years (Bergstrom *et al* 2009 and Carlson 2012).

Accurate estimates on the total escapement of Chum Salmon within the Koyukuk River drainage do not exist. Currently, information is limited to escapement projects on

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tributaries and aerial surveys (McKenna 2014 and Carlson 2014). Currently, there are two escapement projects in the Koyukuk Sub-Basin: Gisasa River and Henshaw Creek (Figure 1). Additionally, data exists from past escapement projects on Clear Creek, the Kateel River, and the South Fork of the Koyukuk River (Dupuis 2012). However, our ability to make accurate inferences from these data to the Koyukuk River stock as a whole are limited due to the short duration of the individual projects or their overall success. In general, Chum Salmon have been documented in 12 Koyukuk tributaries, not including the tributaries that have had escapement projects on them (Barton 1984). Aerial survey records on the Koyukuk River and its tributaries date back to 1960 (Barton 1984). While these data provide a relative index of run strength and not true abundance, aerial survey efforts have increased over the past several years in an attempt to detect inter-annual changes in Chum Salmon production. Based on current aerial surveys, the largest contributors to the production of Chum Salmon to the Koyukuk River include the: Gisasa River, Hogzata River, Henshaw Creek, and Dakli/Wheeler Creek. Together, these rivers and creeks account for roughly 80% of the spawning Chum Salmon counted from the air (Alaska Department of Fish and Game 2016a). However, the accuracy of aerial surveys varies dramatically due to water clarity, substrate color, stream width, weather, aircraft type, observer experience, and water level (Barton 1984). While aerial survey counts may provide an index for river-wide inter-annual run strength, the aforementioned variables make it inaccurate to make inferences on the distribution of Chum Salmon within the Koyukuk River Sub-Basin.

Historically, Chum Salmon pass the Pilot Station Sonar project in the first half of June (Carroll *et al* 2008, JTC 2015) with the e run continuing until mid-July. Spencer and Eiler (2005) found that Koyukuk River stocks pass primarily in the first half of this period. Furthermore, genetic analyses have shown that the upper Koyukuk/Middle Yukon River stocks contributed up to 30% of the first stratum in 2010 (Flannery and Wenburg 2012). During times of higher Chinook Salmon abundance, the first half of the Chum Salmon run will coincide with high subsistence fishing pressure, suggesting that subsistence fishermen rely on Koyukuk River stocks to meet subsistence needs from the mouth of the Yukon River to the villages on the Koyukuk River (Spencer and Eiler 2005). Further, Koyukuk River Chum Salmon may be harvested by commercial fisherman on years with early openings, exposing them to higher exploitation.

In 2012, staff from the Koyukuk National Wildlife Refuge initiated a study to better understand Koyukuk River Chum Salmon migration rates, run timing, relative abundances, and potential tagging bias. Although the Koyukuk River receives a run of both Summer and Fall Chum Salmon (Wiswar 1997), this study has focused on Summer Chum Salmon (i.e., Chum Salmon). The two weirs counting Chum Salmon (Gisasa River and Henshaw Creek weirs) provide a recapture platform for tagged fish; this allows for marked to unmarked ratio data to be collected, providing the base for an abundance estimate.

Optimal tagging locations and run timing at tagging locations were assessed by implanting and tracking spaghetti tags into Chum Salmon in the lower Koyukuk River and recovering at the two aforementioned fish weirs. In 2012, a total of 426 Chum Salmon were tagged and 28 were recovered at the weirs (Gisasa River = 15, Henshaw Creek = 13). Sixty three percent of the tagged fish were females. Average travel time from the tagging locations was 1.6 days (27 km/day) to the Gisasa River weir and 14 days (51.5 km/day) to the Henshaw Creek weir.

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Data from the pilot study were used to set practical objectives for the telemetry project occurring 2013-2016. Ultimately, information from these projects will give researchers baseline information on stocks of Chum Salmon that can be used to develop more comprehensive management and research plans for the entire Yukon River Basin.

Objectives

1. Use radio telemetry to estimate the proportional distribution of Chum Salmon in the Koyukuk River drainage with 95% confidence that the estimate is within +/- 10% of true proportion.
2. Use radio telemetry to detect the ultimate spawning destination upstream of a tagging location (38 rkm), via the presence of at least two tagged fish, of a population comprising 2.5% or more of all the Chum Salmon passing the capture site during each temporal stratum.
3. Describe migration rates and run timing in the Koyukuk River.
4. Identify and document previously unknown Chum Salmon spawning locations.
5. Estimate the abundance of Chum Salmon entering the Koyukuk River with desired precision such that a 90% confidence interval is within +/- 20% of the estimate.

Methods

Project Area

The main-stem of the Koyukuk River begins at the juncture of its North Fork and Middle Fork; north of the Jack White Mountain Range and northeast of Bettles (67.04689° N, 151.07671° W). From there, the Koyukuk flows southwest for 682 rkm where it joins the Yukon River near the village of Koyukuk (Figure 1). The Koyukuk River has a watershed area of 81,327 km² (National Hydrography database, 2011), commonly floods in the spring, and drops to its lowest levels in the fall. Climate of the region is continental subarctic with large seasonal variations in temperature and low annual precipitation. Winter temperatures can drop as low as -56° Celsius while summer temperatures can reach above 27° Celsius. The river usually begins to freeze in October and is free flowing again in May (USFWS, 2009).

Project Design

Capture and Handling Techniques

Capture efforts occurred approximately 30 km upriver from the mouth of the Koyukuk River beginning mid-June and continuing through July. Chum Salmon captures occurred near a spot locally known as Ella's cabin, (approximately 65 km downstream of the Gisasa River). Drift gill nets (10.2 cm x 18.3m x 3m) were deployed from a boat with a two person crew, similar to methods used on the Holitna, Kenai, Kasilof, and Kuskowim Rivers (Chythlook and Evenson, 2003; Carlon and Evans 2007; Gates *et al* ., 2010; and Stuby, 2007), in locations determined during the 2012 and 2013 pilot seasons. Nets were deployed from the bow of the boat with one person piloting and one crewmember tending the net. Drifts were fished until a fish was captured, or until the end of the drifting area was reached. When a fish was detected in the net, the net was pulled, and the portion of the net holding the fish was submerged into a tote holding water. Non-target species were removed from the net and released immediately. Target species were untangled, identified to sex, and measured (mid eye to fork length) using a submerged cradle to reduce stress on the fish. Chum Salmon that showed minimal impacts from capture received tags (spaghetti and/or radio) following tagging procedures described in the appropriate section below. Fish were released on the opposite side of the river from the netting locations to avoid recapture. Total handling time was estimated to be less than 2 minutes per individual. Sampling occurred five days per week, starting at 9:00 a.m. each day.

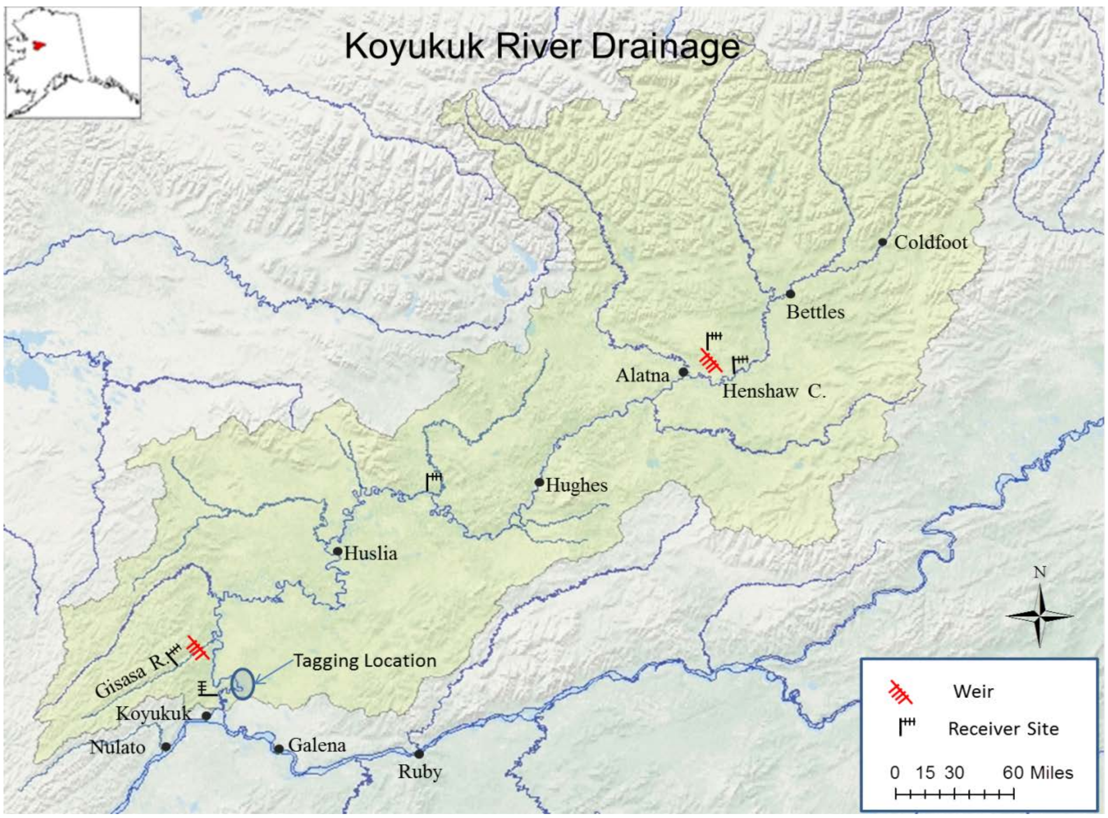

FIGURE 1. — The lower portion of the Koyukuk River Drainage showing tagging location, receiver and weir sites, major rivers and local villages.

Tagging efforts were standardized with a total soak time of 105 minutes per boat per day. A stopwatch was used to time the duration of each set, with timing starting when the net hit the water and ending when the entire net was removed. All healthy Chum Salmon were tagged with a spaghetti tag and every tenth Chum Salmon received a radio transmitter. Each uniquely numbered spaghetti tag was constructed of a section of plastic tubing shrunk onto a piece of monofilament fishing line and sewn through the musculature 1-2 cm ventral of the dorsal fin, between the third and fourth fin ray from the posterior end of the dorsal fin (Stuby, 2007). Several tagging locations from each side of the river were selected and different colored spaghetti tags were used in each strata (Table 1). All spaghetti tags were imprinted with the Koyukuk National Wildlife Refuge phone number so locally harvested fish could be reported.

TABLE 1.—Estimated tag deployment schedule for Summer Chum Salmon telemetry project, Koyukuk River, Alaska, 2014.

Strata	Dates	Radio Transmitters	Spaghetti Tag Colors
1	6/20-6/26	7	Red
2	6/27-7/3	37	Flourescence Yellow
3	7/4-7/10	65	Pink
4	7/11-7/17	66	Flourescence Green
5	7/18-7/24	38	Flourescence Orange
6	7/25-7/31	7	Blue

All capture and handling techniques were designed to prevent or minimize injury and conform to the American Fisheries Society's Guidelines for use of Fish in Field Research (2004).

Tracking Techniques

Advanced Telemetry Systems (ATS; Isanti, Minnesota) model F1840B (58mm x17mm, 22 grams in air) radio transmitters were used to tag adult Chum Salmon following esophageal implantation techniques described by Chythlook and Evenson (2003) and Gates *et al* (2010). Radio transmitters were inserted into the stomach via a polyvinyl chloride (PVC) tube with an internal diameter of 17 mm. The transmitter was pushed through using an 8 mm diameter plunger.

Fish movement was monitored by six fixed station receivers, drainage wide telemetry flights, and periodic tracking via boat. The fixed station receivers were placed in key locations to allow for estimates of travel time to major tributaries as fish traveled up the Koyukuk River. Three tributaries, the Gisasa River, Billy Hawk and Henshaw creeks, had fixed station receivers located in their lower reaches. Three receivers were also set up on the Koyukuk River and placed at Ella's cabin below the tagging sites, the mouth of the Hogatza River, and on the Koyukuk River about 9rkm upstream of the Henshaw Creek (Figure 1).

Fixed station receiver locations included one gel 12-volt deep cycle battery, charged by an 80-watt solar panel and two four element Yagi antennas. An ATS 4100 receiver was used to decode radio transmitter signals, and data was stored in an ATS DCCD5041 data logger. Receivers were programmed to scan through the deployed frequencies at five second intervals. Receiver data was downloaded during aerial telemetry flights.

Aerial surveys employed active tracking techniques where the observer marked a waypoint and noted time and location of decoded transmitters on a data sheet. Aerial surveys used ATS R4500 combination receiver/loggers, which obtained individual radio transmitter codes faster than the older models. Staff from the Kanuti NWR flew one upriver survey in July with an observer from the Fairbanks Fish and Wildlife Field Office. Staff from the Koyukuk NWR flew the remainder of the surveys in July and August.

Radio telemetry data was downloaded in the field, returned to the office, and sorted by channel and transmitter number to ensure accuracy. Waypoint data was stored as latitude and longitude, collected in WGS 84, and projected in Alaska Albers/NAD 83 using ESRI ArcGIS® software Version 10.0.

Objective 1 Methods

The proportional distribution of Chum Salmon in tributaries of the Koyukuk River was estimated by the use of radio transmitters. Fish were captured and monitored using the techniques described in the *Capture and Handling Technique*s and *Tracking Techniques* sections.

Sample size determination for the total target radio transmitter deployment to estimate spawning proportions was addressed by casting spawning locations as multinomial cells and using calculations as described by Bromaghin (1993) to achieve simultaneous multinomial confidence intervals of desired precision. An alpha level of 0.05 and a confidence interval equal to one-half the width of 0.1 (*d*) were selected in an effort to balance cost while still obtaining sufficient data. Fourteen tributaries were identified from the state of Alaska aerial survey database (Alaska Department of Fish and Game, 2013) as tributaries from the Koyukuk River drainage with documented Chum Salmon spawning populations. The list included: Jim, Dulbi, Gisasa, Hogatza, Koyukuk (north, middle and south fork), Kanuti, Alatna, Huslia, Kateel, and Indian Rivers, along with Dakli/Wheeler and Henshaw Creeks. A worst-case scenario, in which one stream would have 50% of the entire return to the Koyukuk River drainage, was used for sample size estimation:

$$n = \left(\frac{z^2_{(1-(\alpha_i/2))}}{2d_i^2}\right)\left[\pi_i(1-\pi_i) - 2d_i^2 + \sqrt{\pi_i^2(1-\pi_i)^2 - d_i^2[4\pi_i(1-\pi_i)-1]}\right]$$

where, for the ith spawning location, we set $\alpha_i = \alpha/k$ with *k* different multinomial cells, $d_i = 0.10$, $\pi_i = 0.50$, and $z_{(1-(\alpha_i/2))}$ is the standard normal deviate exceeded with probability $1 - \frac{\alpha_i}{2}$ (formula from Bromaghin 1993, Table 1 therein).

Using these parameters, the estimated target sample size was 205 Chum Salmon. However, to lessen the chances that an unknown spawning area would influence the analysis, the total number of streams used to estimate sample size was increased to 20. This increased the sample size by 15, for a total of 220 Chum Salmon.

Objective 1 Data Analysis

The proportions of Chum Salmon populations in each tributary was estimated by dividing the number of transmitters located in an individual tributary by the total number of transmitters that arrived on all spawning grounds. Simultaneous confidence intervals for proportions of fish entering individual tributaries by strata were calculated following the Goodman (1965) approach recommended by Bromaghin (1993) and Cherry (1996; equation from Cherry 1996):

$$\hat{p}_i^+ = B + 2(n_i - 0.5) + \frac{\sqrt{B\left(B + \frac{4(n_i - 0.5)(N - n_i + 0.5)}{N}\right)}}{2(N + B)}$$

$$\hat{p}_i^- = B + 2(n_i - 0.5) - \frac{\sqrt{B\left(B + \frac{4(n_i - 0.5)(N - n_i + 0.5)}{N}\right)}}{2(N + B)}$$

where *i*= 1, 2, . . ., *k* spawning grounds, $\hat{p}_i^+$ and $\hat{p}_i^-$ denoted upper and lower confidence limits for cell proportion, *B* corresponded to the upper critical percentile for a Chi-squared distribution with one degree of freedom for a total simultaneous confidence level of 1-α for *k* groups, n_i was the cell count observed for cell *i*, and *N* was the total number of counts. For example, with four groups and a desired 1-α confidence level of 90%, B = 5.024. Note, the above confidence limit equations apportioned type I error equally across all cells (i.e. α/k for each of *k* cells).

Objective 2 Methods:

Radio telemetry was used to detect the ultimate spawning destination upstream of the tagging location (38 rkm) via the presence of at least two tagged fish of a population comprising 2.5% or more of all the Chum Salmon passing the capture site during each temporal stratum. Tributaries that provided Chum Salmon spawning grounds were identified by fating radio-tagged fish to their spawning grounds.

The necessary sample size for the total number of transmitters to release so that a tributary that made up 2.5% or more of the total run would receive two or more radio transmitters with 95% probability, was determined following binomial probability calculations. The objective assumed that marked fish mixed within the general fish population and that both marked and unmarked fish bound for a particular spawning location had an equal probability of arriving at that location. In order to meet the requirements of *Objective 2*, total transmitter deployments needed to be at least 188 tags. The sample requirement for objective two was a compromise between desired accuracy and budgetary concerns.

Objective 2 Data Analysis:

Fish implanted with radio transmitters were assigned one of five possible fates based on data collected from telemetry surveys (Table 2). Spawning locations were defined by aerial telemetry and considered the farthest upstream location the tag was found, provided the location could support the act of spawning. Spawning habitat was defined by the presence of gravel substrate, increased water clarity, and obvious Chum Salmon mortality in the area from completion of spawning. Fish whose upstream migration stopped short of areas that had spawning habitat were considered dead, or to have regurgitated the tag.

TABLE 2.—List of possible fates for Chum Salmon with radio transmitters in the Koyukuk River, Alaska.

Fate	Description
Spawner	A fish that travelled to a spawning location upstream of tagging location.
Dead/Regurgitated	A fish that did not complete the spawning migration or regurgitated a tag before completing its migration. Determined by tag location and lack of movement between tracking flights.
Harvested	A fish that was harvested in the subsistence fishery.
Back Out	A fish that dropped below the downstream tracking receiver station.
Unknown	A fish that could not be assigned another fate with certainty.

Objective 3 Methods

Migration rates and run timing for Chum Salmon implanted with radio transmitters were estimated with radio telemetry and recaptures at the weirs. Gisasa and Henshaw weir crewmembers monitored for spaghetti tagged Chum Salmon and recorded the recapture date and color of all tags passing upstream of the weir. When possible, crewmembers captured the tagged fish, recorded tag numbers, and inspected for radio transmitters. Fish at Billy Hawk Creek and Hogatza River were relocated using fixed station receivers.

Objective 3 Data Analysis:

Migration rates were calculated by comparing the date and time the fish was tagged to the date and time the fish moved past the farthest upstream receiver that last located the fish. Receivers were located on Gisasa River, Hogatza River, Billy Hawk Creek and Henshaw Creek to gather more detailed migration rates on these four stocks.

Run timing was estimated by assigning a fish's tagging date to one of six 7 day strata. The recapture locations at the weirs, fixed station receivers and aerial telemetry provide information on the final fate assigned to a fish. The proportion of fish from each strata returning to a spawning location was calculated and confidence intervals determined using the same techniques described in *Objective 1*.

Objective 4 Methods:

The Alaska Department of Fish and Game (ADFG) maintains an anadromous waters catalog of all known travel, rearing and spawning locations of anadromous fishes throughout the state (https://www.adfg.alaska.gov/sf/SARR/AWC/index.cfm?ADFG=main.interactive). Once telemetry data were analyzed and all spawning locations determined, locations were compared to the ADFG database. Spawning and traveling locations previously undocumented were nominated to be included in the catalog.

Objective 5 Methods:

Abundance of Koyukuk River Chum Salmon was estimated using the 2015 tagging release data and Gisasa Weir recovery data. We compared two abundance estimate techniques; a Chapman two-occasion Lincoln-Petersen closed population estimator ($\widehat{N_C}$, Seber 2002) and a more elaborate time-stratified Lincoln-Peterson model (Sethi and Tanner 2013). The Chapman estimator is a frequently used abundance estimator; however, it relies on the strong assumption of equal capture probability among individuals and violation of this assumption typically results in negatively biased estimates (Baker 2004; Otis et al. 1979). Systematic capture heterogeneity was a concern for this mark-recapture data as river conditions and in-stream abundances change with the strength of the Chum Salmon run.

The Chapman estimator applies a hypergeometric probability distribution to the number of marked individuals captured in a second sampling session (m_2).Confidence intervals for $\widehat{N_C}$ were constructed using a parametric bootstrap routine in program R (R Core Team 2013) by generating 100,000 bootstrap samples of m_2 from a hypergeometric distribution with parameters equal to those under the point estimate of the Chapman estimator. Simulated m_2 values were used to create a distribution of $\widehat{N_C}$'s and subsequent quantiles of the bootstrapped estimates were used to calculate confidence intervals.

We implemented a time-stratified Lincoln-Petersen model in a Bayesian framework in order to explicitly model tag availability for detection and tag detection separately, thereby accommodating potential capture heterogeneity induced during marking or recapture (Sethi and Tanner 2013; Bonner and Schwarz 2011). Primary features of the model include: *i*) deconstruction of the probability of capture at recovery sites into availability for detection and probability of detection, and *ii*) time stratification of parameters (Sethi and Tanner 2013). The assumptions of the time-stratified Lincoln-Petersen model as applied in this study are as follows: 1) additions to the population between the release and recovery site are not possible, and any mortality or emigration between the release and recovery sites is randomly distributed throughout the marked and unmarked population, 2) tags are not shed or misidentified, 3) all animals (i.e., marked and unmarked) that arrive at a given recovery stratum have the same probability of being detected, 4) all animals that pass the release site in a given stratum have the same probability of arriving at each of recovery strata, and finally 5) fish from the same release strata behave independently with respect to movement and fish in the same recovery strata are detected independently. We fit a single model with six recovery and six release strata, following the Pooled-Simple-Simple model form for the Travel Time-Detection Probability-Strata Abundance, respectively, components of the model. Travel times were calculated based upon the midpoint date of release and recovery strata and were modeled as lognormally distributed; a fixed travel time of 1 day was specified for tag release and tag recovery strata that completely overlapped. All prior distributions were as specified in Sethi and Tanner (2013; Table 1 therein), with two exceptions. First, a narrower prior was selected for the scale parameter for the log normal travel model in order to stabilize estimates: $1/\sigma_\lambda \sim Gamma(shape = 10.0, rate = 10.0)$. Posterior checks indicated that posterior distribution for this parameter was narrower than the prior distribution, and thus presumably informed by the data. Second, the priors for stratum specific abundance at the tagging site were shifted larger: $\log(n_i) \sim Normal(mean = 13.0, variance = 4)$. The model was implemented in WinBUGS (Lunn et al. 2000) from the R environment using package 'R2WinBUGS' (Sturtz et al. 2005). Markov chain Monte Carlo implementation used 3 chains each with 70,000 iterations, a 60,000 burn in period, and a 50 iteration thin rate for a combined total of 600 stored posterior draws. Final model convergence was verified by ensuring key parameters of interest (log Normal travel model mean and scale parameter, stratum specific probability of detection, stratum specific abundance at the tagging site, and total abundance) had Gelman-Brooks-Rubin $\hat{R}$ statistics less than 1.2 (Brooks and Gelman 1998). All code for Objective 5 is provided in Appendix 1.

The time-stratified Lincoln-Peterson estimate for Koyukuk River Chum Salmon abundance encompassed the portion of the run that passed the tagging location while tagging occurred; June 22-July 30. The estimate was further expanded to incorporate the components of the run that passed the tagging location outside this time period. One day lags to account for travel time between the tagging location and the Gisasa weir were asserted to link counts at the two locations; an assumption supported by observation and the mean travel time estimated by the time-stratified Lincoln-Peterson model. Fish enumerated at the weir prior to June 23, the first day a tagged fish could arrive at the weir given the travel time estimate, and those arriving after July 31 were estimated by the detection probabilities derived from the time-stratified Lincoln-Peterson model and directly added to the time-stratified Lincoln-Peterson estimate.

This was not, however, a complete estimate because it was known that weir operations on the Gisasa River likely missed the tails of the run. In 2015 an average of 365 fish passed the weir each day in the last week of weir operations and 287 were observed on the last day, July 31. Further, data collected at the Gisasa River weir from 1994-2014 suggest that in any given year the first day of the run is no earlier than June 13 and that the run is complete no later than August 12. To estimate the missing tails, a statistical model as described in Sethi and Bradley (2016) was fit to the weir data that estimated a passage count for each day during the period where no fish count was made at the weir. The model asserts a curve to describe the arrival of fish at the weir with three possible shapes considered: Normal, skew-Normal, and Student's t with two degrees of freedom. Variability about this smooth arrival curve is modeled by Negative Binomial random variables, such that the magnitude of the variability scales positively with the estimated daily passage count. Finally, three correlation structures were proposed for the Negative Binomial probability model: none or white noise, lag-1 autoregressive and lag-1 moving average.

All combinations of run shape and process variation structure were fit to the data, resulting in a candidate set of nine models. Model fits were made in a Bayesian framework, allowing predicted estimates for each missing passage date and the associated variability to be easily obtained. Prior distributions were specified as in Sethi and Bradley (2016, Table 1) and models were run in WinBUGS (Lunn et al. 2000) using R2WinBUGS (Sturtz et al. 2005). Two chains were run with a burn-in period of 1,000 iterations and a thin rate of 10, resulting in 2,000 posterior draws stored. For each model, convergence was ensured by confirming that the Gelman-Brooks-Rubin $\hat{R}$ statistic for the estimated total population abundance was less than 1.2. Deviance information criteria was used to select the best model from the candidate set. Predicted passage from the initial and concluding tails of the run were summed separately and reported as derived parameters.

The total abundance was estimated by expanding the statistical model estimates for the tails by the first and last strata detection probabilities, respectively, and adding these to the time-stratified Lincoln-Peterson model and weir count estimates. Variance estimates were expanded by detection probabilities and summed following the known properties of variances.

Results

A total of 1,765 Chum Salmon were captured between June 22 and July 30, 2015, with 1,330 fish receiving spaghetti tags and 249 fish receiving an additional radio transmitter (approximately every 7th Chum Salmon captured, Table 3, Appendix 2). Other species captured included Chinook Salmon (*O.tshawytscha*, n=2), Broad White fish (*Coregonus nasus*, n=2), Humpback Whitefish (*C. pidschian*, n=1), Northern Pike (*Esox lucius*, n=24), and Sheefish (*Stenodus leucichthys*, n=3) (Table 4).

Soak time totaled 6,300 minutes through 6 strata. The third stratum had the highest Chum Salmon catch rates of the season (n=666) and the maximum deployed spaghetti tags and radio transmitters during that period (Table 3). Capture of non-target species varied by strata with the majority of bycatch occurring during the first four strata (Table 4). A number of fish from each stratum were released without being sampled or tagged due to high catch rates and insufficient holding area (n=435).

TABLE 3.—Total number of Chum Salmon captured and tagged by strata from the lower Koyukuk River, Alaska, 2015.

Strata	Dates	Total Catch	Spaghetti Tag	Radio Transmitter	Not Sampled
1	6/21-6/27	194	157	30	37
2	6/28-7/4	289	228	47	62
3	7/5-7/11	666	491	102	174
4	7/12-7/18	329	240	40	89
5	7/29-7/25	179	145	21	35
6	7/26-7/30	108	69	9	38
Total		1765	1330	249	435

TABLE 4.—Incidentally captured species in the lower Koyukuk River, Alaska, 2015.

		Species				
Strata	Dates	King Salmon	Broad Whitefish	Humpback Whitefish	Northern Pike	Sheefish
1	6/21-6/27	1	1	0	4	2
2	6/28-7/4	0	0	1	5	1
3	7/5-7/11	1	0	0	3	0
4	7/12-7/18	0	0	0	7	0
5	7/19-7/25	0	1	0	4	0
6	7/26-7/30	0	0	0	1	0
	Total	2	2	1	24	3

The overall sex percentage for Chum Salmon across all strata was 36.1% males (n=634) and 63.9% females (n=1,125). Females comprised the majority of samples in all 6 strata (Figure 2). Sex ratios were similar to the overall average during the second stratum (Figure 2). The average length of male Chum Salmon was 567 mm (n=505) and female

Chum Salmon was 533 mm (n=839). Length of both sexes decreased throughout the duration of the season. (Figure 3).

Tagged fish were recaptured at the Gisasa River and Henshaw Creek weirs throughout their operational period. Technicians at the Gisasa River weir counted 64 spaghetti tagged fish between June 23 and July 31, 2015. The Henshaw Creek weir crew recorded 74 spaghetti tagged fish between July 05 and August 04, 2015.

Aerial surveys were conducted on the lower portion of the Koyukuk River (mouth to Hogatza River) during July 26-27 and August 03-04. Upper Koyukuk River (Hogatza River and upstream) were flown July 20-21, and August 12-15. Boat surveys were conducted between the tagging location and the Gisasa River periodically to monitor for fish mortality and regurgitated transmitters.

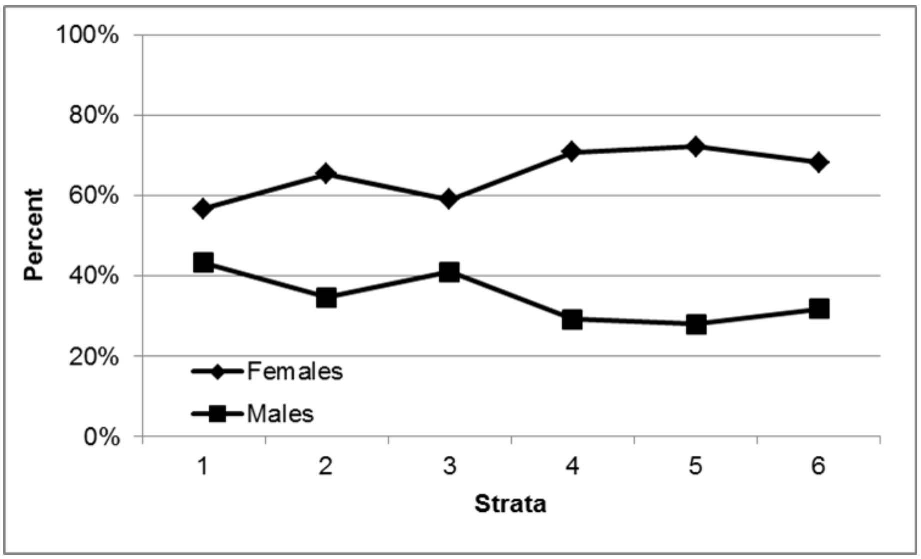

FIGURE 2.—Sex ratios of Chum Salmon captured in the Koyukuk River, 2015.

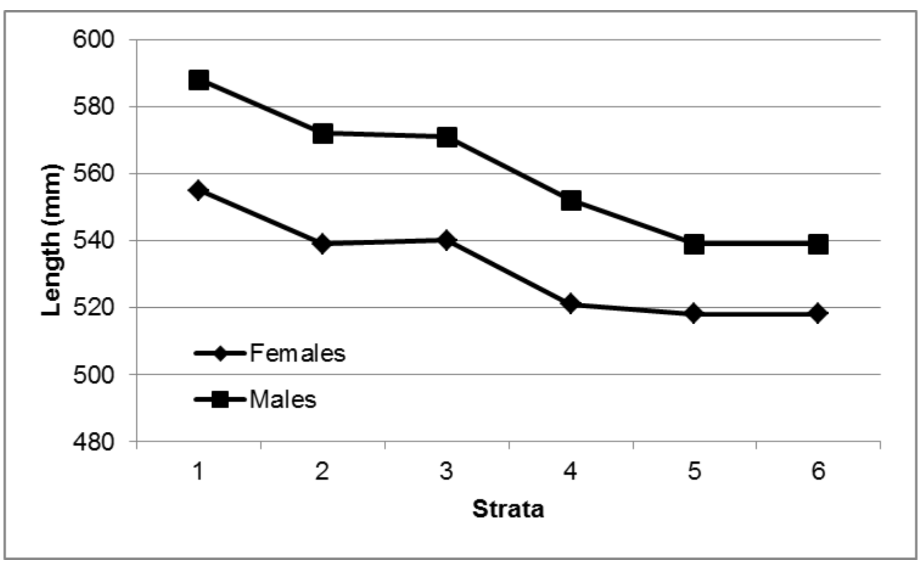

FIGURE 3.—Average length of Chum Salmon captured in the Koyukuk River, 2015.

Aerial surveys covered the mainstream of the Koyukuk River drainage from the mouth up to and including the North, South, and Middle Forks of the River. Surveys encompassed 20 tributaries and recaptures occurred in 10 of the 20 tributaries (Appendix 3). The Gisasa (n=17), Kateel (n=5), Dulbi (2), Huslia/Billy Hawk Creek (n=59), Hogatza (n=19), Indian (n=4), and Alatna Rivers (n=2) along with Dakli Wheeler (n=22), Hughes (n=1), and Henshaw (n=26) Creeks all had fish with transmitters located within the watershed (Table 5, Appendix 3). In addition, 49 fish were suspected of spawning in the mainstream of the Koyukuk River primarily between Treat Island, upstream of Dakli/Wheeler Creek, and Huggins Island near the mouth of the Indian River (Appendix 3). No fish were detected upstream of Henshaw Creek or in the Kanuti River. Eighty three percent of fish implanted with transmitters were tracked to spawning locations, and fish not tracked to spawning locations were classified as dead/regurgitated (n=10, 4%), backed out (n=0, 0%), harvested (n=4, 2%), or unknown (n=30, 12%).

TABLE 5.—Total and proportion of Chum Salmon relocated in the Koyukuk River Drainage, Alaska, 2015. Tributaries are listed in order of occurrence along the Koyukuk River, beginning with the furthest downstream tributary. Distances are from the tagging location to the mouth of each river or creek.

Location	Distance from Tagging Area (rkm)	Radio Transmitters	Proportion	90% CI Upper	90% CI Lower
Gisasa R.	55	17	0.082	0.147	0.045
Kateel R.	93	5	0.024	0.071	0.008
Dulbi R.	185	2	0.010	0.049	0.002
HusliaR/ Billyhawk Cr.	280	59	0.285	0.374	0.212
Dakli/Wheeler Cr.	344	22	0.106	0.176	0.063
Hogatza R.	428	19	0.092	0.159	0.052
Indian R.	542	4	0.019	0.064	0.006
Hughes Cr.	570	1	0.005	0.041	0.001
Alatna R.	697	2	0.010	0.049	0.002
Henshaw Cr.	731	27	0.130	0.203	0.081
Koyukuk R.	477 to 555	49	0.237	0.323	0.170
Total		207	1.000		

Overall, Chum Salmon from the lower Koyukuk River drainage (below the mouth of Hogatza River) were more prevalent throughout the entire run, totaling over 50% of fish tracked to spawning tributaries. At no time during the sampling period were Chum Salmon from the upper Koyukuk River drainage (Hogatza River and upstream) more prevalent (Table 6; Appendix 3). Chum Salmon spawning in the mainstem of the Koyukuk River encompass an area that includes both downstream and upstream areas and were counted separately.

Run timing for most Chum Salmon stocks at the tagging location roughly followed abundance trends at this location, with most stocks abundance peaking in stratum 3 (Table 6). The sole exception being Henshaw Creek, in which 59% of the fish were tagged through strata 2. However, most tributaries had higher percentages of fish tagged in the first three strata than the last three strata. Gisasa River and the Koyukuk River spawning component followed the abundance trends closely, having similar counts of fish in the first three strata and the last three strata.

TABLE 6.—Total and proportion of Chum Salmon recaptured at assumed final spawning locations by strata on the Koyukuk River Drainage, Alaska, 2015. Tributaries are listed in order of occurrence along the Koyukuk River, beginning with the furthest downstream tributary.

	Location in	Total Recapture (Proportion)					
Spawning Location	Drainage	Strata 1	Strata 2	Strata 3	Strata 4	Strata5	Strata 6
Gisasa R.	Lower	2 (.12)	1 (.06)	6 (.35)	4 (.24)	3 (.18)	1 (.06)
Kateel R.	Lower	0 (0)	1 (.20)	2 (.40)	1 (.20)	1 (.20)	0 (0)
Dulbi R.	Lower	0 (0)	0 (0)	1 (.5)	1 (.50)	0 (0)	0 (0)
Huslia/Billy Hawk Cr.	Lower	9 (.15)	17 (.29)	21 (.36)	12 (.20)	0 (0)	0 (0)
Dakli/Wheeler Cr.	Lower	5 (.23)	4 (.18)	9 (.41)	1 (.05)	2 (.09)	1 (.05)
Hogatza R.	Upper	4 (.21)	3 (.16)	11 (.58)	0 (0)	1 (.05)	0 (0)
Indian R.	Upper	0 (0)	1 (.25)	3 (.75)	0 (0)	0 (0)	0 (0)
Hughes Cr.	Upper	0 (0)	1 (1.0)	0 (0)	0 (0)	0 (0)	0 (0)
Alatna R.	Upper	0 (0)	0 (0)	2 (1.0)	0 (0)	0 (0)	0 (0)
Henshaw Cr.	Upper	7 (.26)	9 (.33)	6 (.22)	2 (.07)	3 (.11)	0 (0)
Koyukuk R.		2 (.04)	3 (.06)	23 (.47)	12 (.24)	6 (.12)	3 (.06)
Total		29	40	84	33	16	5

Fish traveling to Gisasa River traveled the 60 rkm in 2 days, while fish returning to Henshaw Creek required 13 days to travel the 731 rkm from the tagging location to the weir site (Table 7). As a result of differences in migration distances, the average distance traveled each day by the separate Chum Salmon stocks varied dramatically. Fish traveling to Billy Hawk Creek and Hogatza River traveled 46 km/day and 41 km/day faster than fish headed to Gisasa River (Table 7). The slowest swimming stocks had the shortest distance (Gisasa River), while the farthest migrating stocks where the second slowest swimming stock. The fish relocated at Henshaw Creek averaged 731 km in 13 days. Fish returning to Billy Hawk Creek and Hogatza river covered the most distance during a day, at 76 rkm and 71 rkm per day respectively. Maybe make a statement about the range/variability? Hogatza range 2x that of other 3, also the lowest # recaps.

TABLE 7.—Travel time from tagging location to the Gisasa River, Billy Hawk Creek, the mouth of the Hogatza River, and Henshaw Creek, Alaska, 2015.

Spawning Location	Number	Distance Traveled (km/day)	Average # Days	Range
Gisasa R.	31	30	2	1-5
Huslia R/ Billyhawk Cr.	55	76	5	4-9
Hogatza R.	19	71	6	5-14
Henshaw Cr.	39	56	13	11-15

The Chapman two-occasion Lincoln-Petersen estimate was 855,485 fish with a 95% confidence interval of {678,144; 1,090,349}, an interval width of approximately +/- 25% of the point estimate. The time-stratified Lincoln-Peterson model indicated increasing tag detection probability over time, with probability of a tag available for detection being successfully detected at the recovery site of 0.02, 0.05, 0.03, 0.06, 0.11, 0.12, across recovery strata 1 through 6, respectively (posterior median values). As expected, when data for closed population estimators exhibit systematic capture heterogeneity, the time-stratified model which corrects for capture heterogeneity produces a larger abundance estimate. The time-stratified estimate of abundance at the tagging site was 931,500 fish with a 95% credibility interval of {717,400; 1,131,000} an interval width of approximately +/- 22% of the point estimate. Finally, the time-stratified model of tag movement indicated a mean travel time of 0.9 days between release and recovery sites, although it should be noted that a fixed travel time of 1 day was specified for tag release and tag recovery strata that completely overlapped (see travel time input data specification in the Appendix 1).

Assuming normality, results from the time-stratified Lincoln-Peterson model described elsewhere provide an estimate of passage from June 22-July 30 at the tagging site of 931,500 with a standard deviation of 99,474. For clarity, we describe this in terms of dates at the Gisasa River weir, i.e., June 23-July 31. Between June 17 and June 22, an additional 365 fish were enumerated at the Gisasa River weir. Using the detection probability estimate from the first stratum of the time-stratified Lincoln-Peterson model, these represent 18,250 fish. Weir operations concluded on July 31; therefore, any estimate of abundance after this date was derived from the statistical model.

To estimate fish passage June 13-16 and August 1- 11, a statistical model with a skew-Normal arrival curve shape and auto-regressive lag-1 serially correlated process variation structure best fit the observed Gisasa River weir data. This model estimated 0 fish (s.d. 1.3) passing the weir prior to June 17 and 2,427 fish (s.d. 1,968; posterior median values) passing the weir after July 31. Expansion of these estimates by their respective detection probabilities and summing the two periods results in an estimate of 20,225 fish (s.d. 16,396) passing the tagging location while the weir was not operational. An abundance estimate inclusive of the total assumed run period is 969,975 with a 95% confidence interval of {772,375; 1,167,575}, an interval width approximately 20% of the point estimate.

Discussion

Excellent water conditions and Chum Salmon abundance throughout the season allowed researchers to meet the minimum requirements for all objectives during 2015. Extremely low water throughout the season likely increased the catch rates during the early and late strata when abundance was low. In 2015, 389 more Chum Salmon were captured, with 161 more of them receiving spaghetti tags than in 2014. In addition, 116 more fish were implanted with radio transmitters during 2015 (Harris *et al.* 2014). The catch rate increased even though the numbers of Chum Salmon counted past the Pilot Station sonar decreased significantly from 2014 (Alaska Department of Fish and Game 2015). In addition, both the Gisasa River and Henshaw Creek weirs were in operation throughout the entire season, providing recapture data that was lacking due to high water in 2014.

The final spawning locations of fish implanted with transmitters showed some trends as they passed the capture locations. Over 80% of the tagged Chum Salmon returning to Henshaw Creek, located 731 rkm upstream from the tagging site, were captured in the first three strata. However, fish returning to Billy Hawk Creek and Gisasa River, located 280 rkm and 55 rkm from the tagging site, were present in larger numbers during the three middle strata. Relatively few fish spawning in the main stem of the Koyukuk River were captured during the first two strata, but fish spawning in the Koyukuk River passed the tagging area in large numbers during the middle two strata. This stock also comprised the majority of tags released during the final two strata. For reference, the Koyukuk River Spawning component primarily straddles the upper/lower river dividing line (Appendix 4).

Salmon that migrate to spawning locations in the upper tributaries of large systems are typically more prevalent in the earlier portions of the run (Eiler *et al.* 2004, Stuby 2007). Although this held true for Henshaw Creek spawning Chum Salmon, no fish were relocated in tributaries upstream of Henshaw Creek in 2015. The fixed station receiver upstream of Henshaw Creek detected only one fish and this fish travelled back downstream and traveled into Henshaw Creek. Chum Salmon have been documented in the Jim River, North, South, and Middle Fork of the Koyukuk River, all of which are upstream of Henshaw Creek, and numbers of fish observed at these locations using aerial fish counting techniques were about average during 2015 (Alaska Department of Fish and Game 2016a). The sole exception was at Henshaw Creek, where numbers were well above average. During a normal year, the numbers of fish observed during aerial surveys in these upstream tributaries are small in comparison to larger spawning areas such as Dakli/Wheeler and Henshaw Creeks. Given these large differences detected by aerial surveys, the possibility exists that the stocks of fish spawning in Jim River, North, South, and Middle Fork of the Koyukuk Rivers were not large contributors to the Koyukuk River Stocks during 2015.

The upper Koyukuk /middle Yukon River stocks typically pass Pilot Station as the first half of the Chum Salmon are counted at the Pilot Station Sonar (Flannery *et al* 2009, Flannery and Evenson 2010, Flannery and Wenberg 2012, 2013, and 2013). However, their timing can vary greatly during that first half of the run. In 2009, the first stratum sampled at Pilot Station contained the highest proportion of upper Koyukuk/Middle Yukon River stocks of any strata, with the proportion dramatically dropping off in the subsequent strata (Flannery and Evenson 2010). Conversely, 2008, 2010, and 2011 all had high proportions of these populations sampled during the first three strata (Flannery *et al* 2009, Flannery and Wenberg 2012 and 2013). The opposite was found in 2012 with high proportions of upper Koyukuk/mainstream Yukon River populations not showing up in the Pilot Stations samples until the last half of the Chum Salmon run (Flannery and Wenberg 2014). Preliminary results from the 2016 Chum Salmon returns past Pilot Station show the upper Koyukuk/mainstream Yukon stocks passing during the middle third of season, during the highest passage of the summer (Blair Flannery, USFWS, personal communications). The possibility exists that fish bound for tributaries in the upper Koyukuk River drainage, including Henshaw Creek, were greatly outnumbered by the lower river stocks as they were passing the tagging area during the peak of the run.

Transmitters were detected throughout the Koyukuk River drainage, except for the extreme upper portions of the Koyukuk River above Henshaw Creek. Conversely, tributaries from the extreme lower portions of the Koyukuk River had high numbers of relocations. Billy Hawk Creek had the largest number of transmitters relocated within the watershed, 54% higher than Henshaw Creek, which had the next, highest number of transmitters relocated in a tributary. This is the second year in which Billy Hawk Creek recorded over 25% of the radio telemetry transmitters deployed during this project (Harris *et al*, 2015). Billy Hawk Creek is a tributary that generally does not have many

Chum Salmon observed during aerial surveys (Alaska Department of Fish and Game, 2013).

The spawning location primarily located on the Koyukuk River between Treat Island and Huggins Island had the second highest number of radio telemetry transmitters relocated on a spawning area. During 2015, low water revealed the gravel substrate in this area. The substrate, along with fish being relocated several times (weeks apart) in a general location without mortality codes being detected provide evidence that this area is being used for spawning. Fish were located in the same areas during 2014 which further supports the likelihood of Chum Salmon spawning in the main stem of the Koyukuk River (Harris *et al*, 2015).

Travel times from the tagging site to the fixed station receivers were faster than what was detected in 2014. Chum Salmon travelling to Gisasa and Hogatza Rivers arrived at their spawning locations 1 and 3 days quicker than recorded in 2014 (Harris *et al*. 2015). An exception was fish swimming to Henshaw Creek, which averaged 13 days in 2014 and 2015. Low water persisted throughout the 2015 season while high water occurred in 2014. River water levels may have allowed the fish to cover more distance in a 24-hour period than in 2014. Limited data is available from Henshaw bound fish in 2014 as only one fish was detected at the fixed station prior to weir removal during flood stage. Also, shallow water at the mouth of Henshaw Creek during 2015 may have delayed entry by fish holding at the mouth awaiting higher water. Chum Salmon spawning in Billy Hawk Creek traveled the fastest, covering 76 km per day, however, there is no swimming speed data from that location in 2014.

Due to low probability of recapture, recaptured fish from the Henshaw Creek weir were not used in the abundance estimates for 2015. Gisasa River weir recorded one spaghetti tagged fish per 668 fish counted through the weir throughout the season, while Henshaw Creek weir recorded approximately one per 3,069 fish counted through the weir throughout the season (Jeremy Carlson, USFWS, personal communications, and Brian McKenna, TCC, personal communications). More interestingly was the variation in the probability of recaptures that Henshaw Creek recorded, with the smallest probabilities occurring during the highest passage periods. This was expected to some extent, but variation in probabilities was not as extreme on the Gisasa River. When using the data from the Henshaw Creek, abundance estimates for the Koyukuk River become very high, rivaling what was counted in the Yukon River with the sonar at Pilot Station. Therefore, we used the data from Gisasa River for 2015 abundance estimates.

Theories as to why the probability of recaptures was low and highly variable in Henshaw Creek are largely the same ideas discussed in the paragraphs above. The fish may have returned early enough that tagging missed a large portion, however genetics information at Pilot Station and run timing at the Henshaw Creek weir present evidence that suggest otherwise (Brian McKenna, TCC, personal communications). Looking at peak passage of Chum salmon through the Henshaw Creek weir and subtracting the 13 day average travel time determined from this telemetry project, we are able to determine that the peak time of travel past the tagging location for fish heading to Henshaw Creek weir was during the busiest week of the season. Thus, the sheer numbers of Chum Salmon passing the tagging location may have reduced the concentration of this stock to the point that they were difficult to capture. Another possibility is that the sheer numbers of fish passing through the weir made it difficult to observe spaghetti tags as fish passed through the counting chute. Technicians counted over 10,000 fish per day during an 8 day stretch from July 17 to July 24, surpassing 22,000 on July 20 (Brian McKenna, TCC, personal communication). Fish generally pass through a weir in pulses (Weber 2013). The high densities of fish coupled with waves of fish passing close together may have made observing spaghetti tags more difficult than in locations with lower densities.

The abundance estimates represents one of the few attempts at determining the number of Chum Salmon that enter the entire Koyukuk River system. The time stratified point estimate represents approximately 70% of the Chum Salmon counted past the ADF&G sonar located at Pilot Station, and over 2.5 times the number counted at the Anvik River Sonar (ADF&G, 2016b). While it is unlikely that the Koyukuk River Chum Salmon stocks represented 70% of the entire Yukon River Chum Salmon return, it is likely that Koyukuk River Chum Salmon are a major producer within the drainage. The Anvik River accounted for less than 30% of the fish counted past the sonar, while Chum Salmon escapement in the Yukon River was slightly below median in 2015 (ADF&G 2016b). Potential differences in the sonar and Koyukuk River Chum Salmon estimates could be the result of the sonar undercounting fish, over estimation from bias in tagging and recaptures from the mark recapture, or a combination of both scenarios.

Recommendations

In order to meet the objective of 220 radio transmitters deployed in 2016, and given a surplus of tags leftover from the 2015 season, it is recommended that the marking rate be started at 1:6 in 2016; depending on preseason forecast and early Pilot Station sonar numbers. The left over transmitters from 2015 along with the 220 new transmitters will allow researchers to spaghetti tag up to 1,550 Chum Salmon prior to running out of available transmitters. This rate would also meet the minimum goal of 220 transmitters deployed with a catch of 1320 Chum Salmon. A similar escapement as what was observed in 2015 should allow researchers to accomplish the minimum number of radio transmitters deployed at the 1:6 marking rate.

In order to confirm that fish are actually spawning in the main stem of the Koyukuk River, researchers should sample this location in early August.

Capture of Chum Salmon should start at least 2 days prior (approximately June 16) to the Gisasa River weir beginning operations. Starting capture at this time will give researchers a better chance of capturing early stocks that may be heading to the farthest up stream tributaries while allowing for a more comprehensive abundance estimate for the drainage.

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APPENDIX 1.— Model code used to estimate the time- stratified Lincoln-Peterson estimate for Koyukuk River Chum Salmon abundance.

```
library(R2WinBUGS)
bd <- "C:\\Program Files (x86)\\winbugs14\\WinBUGS14" # from home
laptop

# 2015 Koy data: Gisasa
s <- 6 # release strata
t <- 6 # recovery strata
m. <- c(157,228,491,240,145,50) # s release strata
c. <- c(1485,6264,10132,15598,6819,2020) # t recovery strata  #THINK
1814 SHOULD BE 1485
R <- matrix(nr=s,nc=(t+1),byrow=T,c(2,4,0,0,0,0,151,
                                    0,4,2,0,0,0,222,
                                    0,0,11,5,0,0,475,
                                    0,0,0,8,5,0,227,
                                    0,0,0,0,14,4,127,
                                    0,0,0,0,0,5,45))

# Gisasa 2015:
# rel str: 1=6/22-6/26; 2=6/29-7/3; 3=7/6-7/10; 4=7/13-7/17; 5=7/20-
7/24; 6=7/27-7/30
# rec str: 1=6/23-6/27; 2=6/28-74; 3=7/5-7/11; 4=7/12-7/18; 5=7/19-
7/25; 6=7/26-7/31
# define difference in days as midpoint of release strata to
midpoint of recovery strata
# Rows are release strata, columns are recovery strata
X <- matrix(nr=s,nc=t,byrow=T,c(1,7,14,21,28,34.5,
                                0.001,1,7,14,21,27.5,
                                0.001,0.001,1,7,14,20.5,
                                0.001,0.001,0.001,1,7,13.5,
                                0.001,0.001,0.001,0.001,1,6.5,
                                0.001,0.001,0.001,0.001,0.001,1))
# note here, I've included "0.001" as a placeholder travel time for
cases where recovery strata begins
# before release occurs.  In the subsequent travel model, lambda,
prob. of availability for detection,
# will be calculated as very small numbers close to zero.

# Traditional LP Confidence interval Method 1: parametric bootstrap
with hypergeometric distribution
n1 <- sum(m.); n2 <- sum(c.)+sum(R[,1:t]); m2 <- sum(R[,1:t])
LP <- (n1+1)*(n2+1)/(m2+1) # 855485
nn.<- 100000 # the number of random draws
mm <- n1 # n1 in the Chapman estimator, the number of marks ("white
balls" by R) in the urn
n. <- LP-n1 # the number of unmarked fish in the urn at time of
recapture
k. <- n2 # the number of balls drawn from the urn, the recapture
sample
m2.sim <- rhyper(nn=nn. , m=mm , n=n., k=k. )
N.sim <- round((n1+1)*(n2+2)/(m2.sim+1)) # bootstrap point estimate
CI.Bootstrap <- quantile(N.sim,probs=c(0.025,0.975)) # bootstrap
confidence interval, 678144 to 1090349, about +/- 25%
```

APPENDIX 1.—Continued

```
# Fit the Pooled (travel) - Simple (detection) - Simple (abundance)
time stratified model
# NOTE: Need to modify travel time model for code to only examine
recaptures for recovery strata >= than release,
# and need to update priors for travel time to reflect shorter
travel time beliefs, and abundance to reflect
# larger abundance beliefs.


win.data <- list(m.=m.,c.=c.,s=s,t=t,R=R,X=X)
# nc = 1; ni = 5000; nb = 400; nt = 2
nc = 3; ni = 70000; nb = 60000; nt = 50

## model P.S.S
inits.PSS <- function(){
  list(LNmu_travel = 0+runif(1,-.2,.2), tau_lambda=1+runif(1,-
.1,.1),
       p.logit=1+runif(t,-.1,.1),
       n.log = 11.5+runif(s,0,1) ) }

params.PSS <- c("tau_lambda", "LNmu_travel",
"LNsigma_travel","lambda",
                "p",
                "n",
                "p_recap", "u", "Ntot", "Ntot_u")
out.PSS <-  bugs(data=win.data,inits=inits.PSS,
parameters=params.PSS, model = "mod.PSS.Koy.txt", n.thin=nt,
n.chains=nc,
                 n.burnin=nb, n.iter=ni, debug=T, DIC=F,
working.directory=getwd() )

dput(out.PSS,"KoyChum2015.Run4.3.7.16.txt") # Run 1 has LNmu_travel
~ dnorm(0,.2) and i==j 0.5 travel day;
# Run 2 has LNmu_travel ~ dnorm(0,.2) and i==j 1 travel day
# Run 3 has LNmu_travel ~ dnorm(0,.2), tau_lambda ~ dgamma(10,10)
and i==j 1 travel day
# Run 4: as Run 3 but with 3 chians 700k iter, 600k burn, 50 thin =
3 x 2000 = 6000 posterior draws
# quick check on posterior for travel sigma:
hist(1/rgamma(n=nc*((ni-
nb)/nt),10,10),breaks=50,col="gray75",freq=F)
hist(out.PSS$sims.matrix[,"LNsigma_travel"],breaks=50,col="gray45",f
req=F,add=T)

##### Model Code
### Model P.S.S version Koyukuk
sink("mod.PSS.Koy.txt")
cat("
    model {

    ## Pooled model ##
    # priors
    LNmu_travel ~ dnorm(0,0.25)
```

APPENDIX 1.—Continued

```
    tau_lambda ~ dgamma(10,10) # equal to 1/(sigma_lambda) as
defined here (so not precision, but sqrt precision)
    LNsigma_travel <- 1/tau_lambda


   # calculate lambda probabilities using WinBUGS cumulative Normal
distribution function on standardized travel times
    # lambda[i,j] = probability a marked released in i arrives to be
available for capture in j
    for(i in 1:s){
    lambda[i,1] <- phi((log(X[i,1])-LNmu_travel)/LNsigma_travel) #
for first recovery strata
    for(j in 2:t){
    lambda[i,j] <- phi((log(X[i,j])-LNmu_travel)/LNsigma_travel)-
phi((log(X[i,j-1])-LNmu_travel)/LNsigma_travel)
    } # end j loop
    # fish never seen again -- not used in likelihood, but can be
tracked if desired
    lambda[i,t+1] <- 1-sum(lambda[i,1:t])
    } # end i loop

    ## Simple p model ##
    # priors
    for (j in 1:t){
    p.logit[j] ~ dnorm(0,.3333)
    p[j] <- exp(p.logit[j])/(1+exp(p.logit[j]))
    } # end j loop

    ## Simple n model ##
    # priors
    for (i in 1:s){
    n.log[i]~dnorm(13,.25)
    log(n[i]) <- n.log[i]
    } # end j loop

    ### Likelihoods
    ## Contribution from marks (equation XX in main text)
    for(i in 1:s){
    for(j in 1:t){
    p_recap[i,j] <- lambda[i,j]*p[j]
    } # end j loop
    p_recap[i,t+1] <- 1 - sum(p_recap[i,1:t]) # the probability of a
mark released in i never being recovered again.
    # Multinomial likelihood contribution from releases in i
    R[i,1:(t+1)] ~ dmulti(p_recap[i,],m.[i])
    } # end i loop

    ## Contribution from unmarks (equation XX in main text)
    for(j in 1:t){
    for (i in 1:s){
    store.u[j,i] <- n[i]*lambda[i,j] # n[1]*lambda[1,j] + ... +
n[s]*lambda[s,j]
    } # end i loop
```

APPENDIX 1.—Continued

```
    u[j]<-round(sum(store.u[j,1:s]))
    c.[j] ~ dbin(p[j],u[j])
    } # end j loop

    ### Derived parameter of interest: total run size
    Ntot_u <- sum(u[1:t])+sum(m.[1:s])
    Ntot <- sum(n[1:s]) + sum(m.[1:s])
    } # end winBUGS model
    ",fill=T)
sink()
### End Model 10: P.S.S
```

APPENDIX 2.— Summary of biological data and assigned fate of Chum Salmon implanted with radio transmitters in the Koyukuk River, Alaska, 2015.

Tagging Stratum	Date Tagged	Sex	Length	Spaghetti Tag #	Radio Frequency	Radio Tag #	Fate
1	6/22/15	M	620	3390	162.123	35	Hogatza R.
1	6/22/15	M	550	3385	162.183	57	Gisasa R.
1	6/23/15	F	575	3400	162.573	50	Henshaw Cr.
1	6/23/15	M	585	3394	162.234	43	Billy Hawk Cr.
1	6/23/15	M	600	126	162.334	39	Gisasa R.
1	6/23/15	M	640	3388	162.134	53	Hogatza R.
1	6/24/15	M	590	147	162.573	39	Billy Hawk Cr.
1	6/24/15	F	570	146	162.334	42	Koyukuk R.
1	6/24/15	M	620	131	162.224	55	Captured
1	6/24/15	M	590	94	162.224	49	Dakli/Wheeler Cr.
1	6/24/15	F	565	84	162.123	58	Dead/Regurgitated
1	6/24/15	F	570	88	162.234	44	Billy Hawk Cr.
1	6/24/15	F	560	249	162.234	53	Henshaw Cr.
1	6/24/15	F	590	81	162.224	45	Billy Hawk Cr.
1	6/24/15	F	590	229	162.183	41	Billy Hawk Cr.
1	6/24/15	F	565	279	162.123	43	Dakli/Wheeler Cr.
1	6/24/15	M	640	278	162.224	58	Billy Hawk Cr.
1	6/24/15	M	580	233	162.573	43	Hogatza R.
1	6/25/15	F	570	203	162.174	41	Billy Hawk Cr.
1	6/25/15	M	585	218	162.123	33	Henshaw Cr.
1	6/25/15	M	570	212	162.183	38	Billy Hawk Cr.
1	6/25/15	F	595		162.174	52	Koyukuk R.
1	6/25/15	M	590	192	162.334	34	Billy Hawk Cr.
1	6/25/15	M	585	198	162.183	42	Henshaw Cr.
1	6/25/15	F	610	179	162.334	56	Dakli/Wheeler Cr.
1	6/25/15	M	585	204	162.174	56	Dakli/Wheeler Cr.
1	6/25/15	M	600	240	162.183	47	Henshaw Cr.
1	6/25/15	F	590	285	162.334	49	Dakli/Wheeler Cr.
1	6/25/15	M	620	120	162.224	51	Henshaw Cr.
1	6/25/15	F	560	286	162.573	34	Henshaw Cr.
1	6/25/15	M	570	295	162.234	35	Hogatza R.
2	6/29/15	M	575	3511	162.183	50	Billy Hawk Cr.
2	6/29/15	M	560	984	162.134	37	Gisasa R.
2	6/29/15	F	590	995	162.123	54	Billy Hawk Cr.
2	6/29/15	M	595	477	162.123	40	Henshaw Cr.
2	6/29/15	M	615	734	162.234	47	Billy Hawk Cr.
2	6/30/15	M	585	3521	162.123	39	Hogatza R.
2	6/30/15	M	615	3509	162.134	47	Henshaw Cr.
2	6/30/15	F	625	3507	162.123	56	Unknown
2	6/30/15	F	565	3723	162.334	33	Billy Hawk Cr.
2	6/30/15	M	555	3720	162.573	42	Kateel R.
2	6/30/15	F	575	3516	162.134	46	Henshaw Cr.
2	6/30/15	M	570	726	162.224	46	Henshaw Cr.
2	6/30/15	F	550	996	162.134	54	Billy Hawk Cr.
2	6/30/15	F	550	990	162.123	50	Henshaw Cr.

APPENDIX 2.— Continued .

Tagging Stratum	Date Tagged	Sex	Length	Spaghetti Tag #	Radio Frequency	Radio Tag #	Fate
2	6/30/15	F	560	730	162.183	39	Henshaw Cr.
2	6/30/15	M	590	932	162.224	38	Koyukuk R.
2	7/1/15	F	580	3718	162.224	44	Henshaw Cr.
2	7/1/15	F	550	3704	162.224	39	Billy Hawk Cr.
2	7/1/15	M	560	3683	162.234	54	Dakli/Wheeler Cr.
2	7/1/15	M	585	3679	162.334	44	Billy Hawk Cr.
2	7/1/15	F	575	3697	162.224	34	Billy Hawk Cr.
2	7/1/15	F	590	3659	162.334	40	Dead/Regurgitated
2	7/1/15	M	575	3671	162.183	58	Dakli/Wheeler Cr.
2	7/1/15	M	565	3664	162.234	55	Billy Hawk Cr.
2	7/1/15	F	560	3672	162.174	57	Dakli/Wheeler Cr.
2	7/1/15	F	560	3654	162.234	56	Billy Hawk Cr.
2	7/1/15	M	565	3667	162.183	56	Indian R.
2	7/1/15	F	575	3674	162.123	45	Billy Hawk Cr.
2	7/1/15	M	575	3660	162.174	44	Unknown
2	7/2/15	F	570	3646	162.174	33	Koyukuk R.
2	7/2/15	F	560	3633	162.123	53	Billy Hawk Cr.
2	7/2/15	F	560	3643	162.234	49	Koyukuk R.
2	7/2/15	F	570	938	162.174	38	Dead/Regurgitated
2	7/2/15	M	620	929	162.224	54	Henshaw Cr.
2	7/2/15	F	565	943	162.334	55	Billy Hawk Cr.
2	7/2/15	F	550	936	162.234	37	Hogatza R.
2	7/3/15	M	620	3635	162.334	43	Billy Hawk Cr.
2	7/3/15	M	555	3616	162.183	55	Captured
2	7/3/15	F	580	3617	162.123	51	Unknown
2	7/3/15	F	575	920	162.573	51	Billy Hawk Cr.
2	7/3/15	M	650	968	162.234	57	Hughes Cr.
2	7/3/15	M	600	959	162.234	38	Dakli/Wheeler Cr.
2	7/3/15	M	590	956	162.183	32	Billy Hawk Cr.
2	7/3/15	M	565	961	162.134	36	Billy Hawk Cr.
2	7/3/15	M	600	973	162.174	48	Henshaw Cr.
2	7/3/15	M	585	969	162.224	47	Hogatza R.
3	7/6/15	M	620	1449	162.183	54	Koyukuk R.
3	7/6/15	M	560	1787	162.224	52	Dakli/Wheeler Cr.
3	7/6/15	F	575	1423	162.224	56	Unknown
3	7/6/15	F	560	1431	162.234	32	Dead/Regurgitated
3	7/6/15	M	590	3882	162.234	46	Dakli/Wheeler Cr.
3	7/6/15	M	565	1435	162.334	45	Billy Hawk Cr.
3	7/6/15	M	580	1790	162.174	40	Koyukuk R.
3	7/6/15	M	560	1798	162.174	31	Billy Hawk Cr.
3	7/6/15	M	580	1441	162.174	37	Dakli/Wheeler Cr.
3	7/6/15	M	600	1409	162.174	45	Unknown
3	7/6/15	M	605	3878	162.334	47	Kateel R.
3	7/6/15	M	590	1445	162.573	45	Billy Hawk Cr.
3	7/6/15	F	555	3885	162.573	46	Billy Hawk Cr.

APPENDIX 2.— Continued.

Tagging Stratum	Date Tagged	Sex	Length	Spaghetti Tag #	Radio Frequency	Radio Tag #	Fate
3	7/7/15	F	590	4112	162.123	47	Hogatza R.
3	7/7/15	F	620	1414	162.183	51	Koyukuk R.
3	7/7/15	M	590	1489	162.183	44	Dead/Regurgitated
3	7/7/15	M	620	1208	162.183	35	Dead/Regurgitated
3	7/7/15	F	570	1415	162.224	43	Billy Hawk Cr.
3	7/7/15	M	640	4120	162.234	41	Henshaw Cr.
3	7/7/15	F	560	1494	162.123	36	Unknown
3	7/7/15	M	595	1498	162.123	49	Henshaw Cr.
3	7/7/15	M	550	4188	162.224	42	Hogatza R.
3	7/7/15	M	645	1402	162.134	45	Koyukuk R.
3	7/7/15	M	600	1486	162.134	41	Koyukuk R.
3	7/7/15	M	585	4179	162.134	42	Dakli/Wheeler Cr.
3	7/7/15	M	570	1419	162.174	34	Unknown
3	7/7/15	M	580	1408	162.174	39	Billy Hawk Cr.
3	7/7/15	M	615	1490	162.174	53	Koyukuk R.
3	7/7/15	F	560	4183	162.174	47	Koyukuk R.
3	7/7/15	M	600	1406	162.334	41	Billy Hawk Cr.
3	7/7/15	F	580	1481	162.334	50	Koyukuk R.
3	7/7/15	M	600	4192	162.334	54	Dakli/Wheeler Cr.
3	7/7/15	M	660	4177	162.573	55	Koyukuk R.
3	7/8/15	F	590	4147	162.174	44	Unknown
3	7/8/15	M	570	4154	162.174	46	Billy Hawk Cr.
3	7/8/15	M	615	1204	162.183	40	Henshaw Cr.
3	7/8/15	M	580	1253	162.183	37	Dakli/Wheeler Cr.
3	7/8/15	F	565	1260	162.183	48	Unknown
3	7/8/15	F	555	4144	162.183	34	Dakli/Wheeler Cr.
3	7/8/15	M	580	1203	162.224	41	Indian R.
3	7/8/15	F	560	1270	162.224	36	Billy Hawk Cr.
3	7/8/15	M	575	1272	162.234	51	Unknown
3	7/8/15	M	610	4106	162.234	39	Alatna R.
3	7/8/15	M	640	4137	162.123	44	Alatna R.
3	7/8/15	M	590	4168	162.123	31	Koyukuk R.
3	7/8/15	F	565	1201	162.134	50	Koyukuk R.
3	7/8/15	F	580	1263	162.134	52	Koyukuk R.
3	7/8/15	M	570	4101	162.134	55	Gisasa R.
3	7/8/15	M	560	4160	162.134	39	Billy Hawk Cr.
3	7/8/15	F	565	1224	162.174	32	Gisasa R.
3	7/8/15	M	600	1256	162.174	54	Indian R.
3	7/8/15	M	570	1288	162.174	43	Billy Hawk Cr.
3	7/8/15	F	560	4141	162.334	52	Koyukuk R.
3	7/8/15	F	560	4145	162.573	35	Koyukuk R.
3	7/8/15	M	550	4170	162.234	52	Hogatza R.
3	7/9/15	M	600	3802	162.183	43	Hogatza R.
3	7/9/15	M	640	3868	162.224	37	Hogatza R.
3	7/9/15	F	585	1297	162.174	58	Koyukuk R.

APPENDIX 2.— Continued.

Tagging Stratum	Date Tagged	Sex	Length	Spaghetti Tag #	Radio Frequency	Radio Tag #	Fate
3	7/9/15	F	560	3823	162.174	36	Captured
3	7/9/15	F	570	3843	162.174	50	Billy Hawk Cr.
3	7/9/15	M	630	1261	162.183	33	Koyukuk R.
3	7/9/15	F	580	1300	162.183	49	Dakli/Wheeler Cr.
3	7/9/15	M	580	1243	162.224	53	Billy Hawk Cr.
3	7/9/15	M	625	1240	162.224	50	Billy Hawk Cr.
3	7/9/15	M	570	3804	162.224	33	Billy Hawk Cr.
3	7/9/15	M	600	3827	162.224	35	Gisasa R.
3	7/9/15	F	580	1291	162.234	45	Billy Hawk Cr.
3	7/9/15	M	595	1229	162.234	34	Henshaw Cr.
3	7/9/15	F	570	3843	162.234	31	Unknown
3	7/9/15	F	570	3822	162.123	34	Kateel R.
3	7/9/15	M	575	3830	162.123	52	Billy Hawk Cr.
3	7/9/15	M	580	3862	162.123	42	Indian R.
3	7/9/15	M	585	1238	162.134	40	Unknown
3	7/9/15	M	610	3860	162.134	35	Koyukuk R.
3	7/9/15	M	580	1245	162.334	46	Billy Hawk Cr.
3	7/9/15	F	590	1551	162.334	31	Unknown
3	7/9/15	M	570	1558	162.334	51	Unknown
3	7/9/15	F	585	3808	162.234	48	Hogatza R.
3	7/9/15	M	580	3851	162.573	33	Gisasa R.
3	7/9/15	M	575	1234	162.573	44	Hogatza R.
3	7/9/15	F	570	1237	162.573	54	Hogatza R.
3	7/10/15	M	590	1507	162.134	51	Hogatza R.
3	7/10/15	F	575	1520	162.123	32	Koyukuk R.
3	7/10/15	M	605	4067	162.123	41	Henshaw Cr.
3	7/10/15	F	565	1553	162.134	44	Billy Hawk Cr.
3	7/10/15	M	570	1573	162.134	38	Dulbi R.
3	7/10/15	F	570	1569	162.134	34	Unknown
3	7/10/15	M	590	1587	162.134	33	Billy Hawk Cr.
3	7/10/15	M	560	4035	162.134	43	Henshaw Cr.
3	7/10/15	M	565	4048	162.134	32	Gisasa R.
3	7/10/15	M	560	4032	162.334	48	Unknown
3	7/10/15	M	565	4031	162.334	53	Koyukuk R.
3	7/10/15	M	575	4045	162.334	32	Gisasa R.
3	7/10/15	F	555	4059	162.334	37	Dead/Regurgitated
3	7/10/15	M	570	1556	162.573	47	Koyukuk R.
3	7/10/15	M	570	1513	162.573	49	Koyukuk R.
3	7/10/15	F			162.573	41	Billy Hawk Cr.
3	7/10/15	M	600	1510	162.573	31	Koyukuk R.
3	7/10/15	M	620	1515	162.573	52	Hogatza R.
3	7/10/15	M	585	4027	162.573	36	Koyukuk R.
3	7/10/15	M	635		162.573	37	Dakli/Wheeler Cr.
3	7/10/15	M	615	1597	162.573	56	Hogatza R.
4	7/13/15	M	580	4215	162.234	42	Unknown

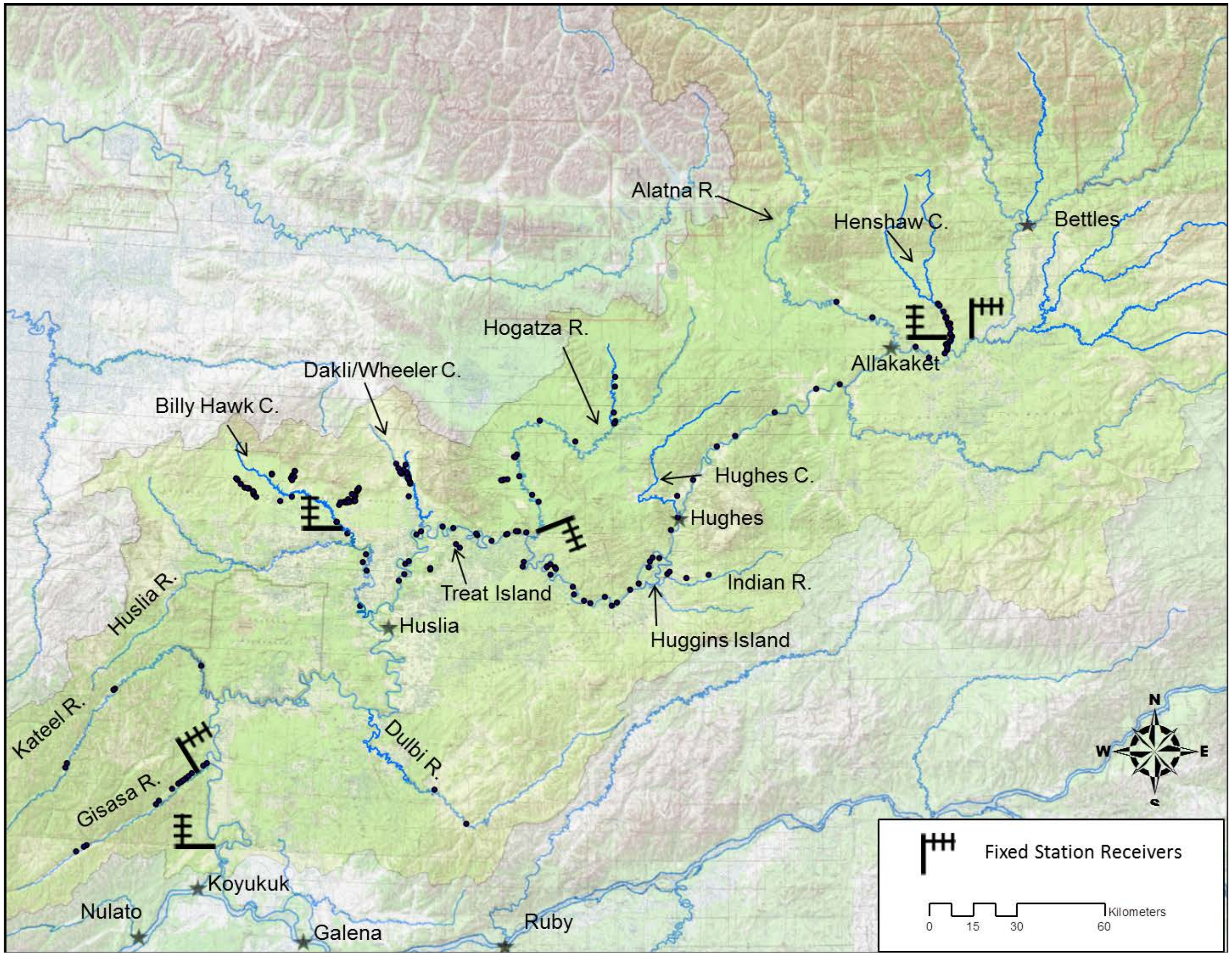
APPENDIX 2.— Continued.

Tagging Stratum	Date Tagged	Sex	Length	Spaghetti Tag #	Radio Frequency	Radio Tag #	Fate
4	7/13/15	M	610	4223	162.573	48	Koyukuk R.
4	7/14/15	F	600	4218	162.123	55	Henshaw Cr.
4	7/14/15	F	575	2062	162.234	36	Koyukuk R.
4	7/14/15	M	625	2054	162.573	57	Gisasa R.
4	7/14/15	M	615	2071	162.123	27	Billy Hawk Cr.
4	7/14/15	M	590	1819	162.183	31	Henshaw Cr.
4	7/15/15	M	620	2061	162.134	58	Koyukuk R.
4	7/15/15	F	580	2091	162.183	45	Unknown
4	7/15/15	F	565	2097	162.123	38	Unknown
4	7/15/15	M	585	2096	162.573	40	Billy Hawk Cr.
4	7/15/15	F	565	2076	162.224	48	Koyukuk R.
4	7/15/15	M	570	2031	162.134	57	Billy Hawk Cr.
4	7/15/15	F	565	2038	162.234	33	Koyukuk R.
4	7/15/15	F	595	2030	162.123	48	Gisasa R.
4	7/15/15	M	570	2032	162.334	38	Billy Hawk Cr.
4	7/15/15	M	615	2034	162.573	53	Unknown
4	7/15/15	F	560	2010	162.174	49	Unknown
4	7/15/15	M	590	1898	162.224	57	Gisasa R.
4	7/15/15	M	605	1807	162.134	56	Koyukuk R.
4	7/15/15	F	575	1810	162.334	35	Billy Hawk Cr.
4	7/15/15	F	580	4586	162.174	35	Koyukuk R.
4	7/15/15	M	610	4592	162.134	31	Dakli/Wheeler Cr.
4	7/16/15	F	560	2077	162.183	52	Billy Hawk Cr.
4	7/16/15	F	590	2040	162.183	53	Koyukuk R.
4	7/16/15	F	580	2029	162.234	50	Koyukuk R.
4	7/16/15	M	560	2148	162.334	36	Gisasa R.
4	7/16/15	F	570	2141	162.174	23	Koyukuk R.
4	7/16/15	F	560	4584	162.573	38	Koyukuk R.
4	7/16/15	M	590	4575	162.123	37	Billy Hawk Cr.
4	7/16/15	F	560	4561	162.174	55	Dulbi R.
4	7/16/15	M	590	4556	162.224	17	Kateel R.
4	7/16/15	F	560	4535	162.183	36	Dead/Regurgitated
4	7/17/15	F	575	2133	162.573	27	Koyukuk R.
4	7/17/15	F	570	2126	162.134	26	Billy Hawk Cr.
4	7/17/15	M	565	2176	162.134	23	Billy Hawk Cr.
4	7/17/15	M	565	2192	162.334	21	Billy Hawk Cr.
4	7/17/15	M	565	2199	162.134	3	Billy Hawk Cr.
4	7/17/15	F	580	2186	162.134	11	Billy Hawk Cr.
4	7/17/15	F	565	4531	162.224	32	Unknown
5	7/20/15	M	560	2530	162.123	46	Koyukuk R.
5	7/20/15	M	570	2522	162.234	75	Henshaw Cr.
5	7/20/15	F	585	2516	162.334	5	Koyukuk R.
5	7/20/15	F	590	2517	162.123	26	Dead/Regurgitated
5	7/20/15	F	575	2508	162.123	43	Dakli/Wheeler Cr.
5	7/20/15				162.123	23	Dakli/Wheeler Cr.

APPENDIX 2.— Continued.

Tagging Stratum	Date Tagged	Sex	Length	Spaghetti Tag #	Radio Frequency	Radio Tag #	Fate
5	7/21/15	F	565	2550	162.183	46	Koyukuk R.
5	7/21/15	M	565	4775	162.573	32	Dead/Regurgitated
5	7/21/15	F	580	4993	162.183	75	Kateel R.
5	7/22/15	M	585	4764	162.134	48	Hogatza R.
5	7/22/15	M	570	4766	162.234	40	Captured
5	7/22/15	M	590	4765	162.224	75	Henshaw Cr.
5	7/22/15	F	560	4761	162.183	14	Unknown
5	7/22/15	M	615	4791	162.183	13	Henshaw Cr.
5	7/22/15	F	580	4773	162.183	18	Gisasa R.
5	7/22/15	M	560	4982	162.174	3	Koyukuk R.
5	7/23/15	M	570	4756	162.183	22	Unknown
5	7/23/15	F	560	4951	162.573	12	Koyukuk R.
5	7/23/15	F	580	4957	162.123	4	Gisasa R.
5	7/23/15	F	590	4967	162.224	13	Unknown
5	7/23/15	F	565	4948	162.234	24	Gisasa R.
5	7/23/15	M	590	4947	162.134	12	Koyukuk R.
6	7/27/15	M	565	3092	162.334	15	Dakli/Wheeler Cr.
6	7/28/15	M	575	3090	162.573	20	Gisasa R.
6	7/29/15	M	615	3048	162.134	1	Unknown
6	7/29/15	M	575	3044	162.134	16	Koyukuk R.
6	7/29/15	F	560	3043	162.334	26	Unknown
6	7/29/15	F	560	3118	162.224	40	Unknown
6	7/29/15	F	570	3105	162.123	57	Koyukuk R.
6	7/30/15	F	565	3041	162.183	16	Unknown
6	7/30/15	M	570	3078	162.224	21	Koyukuk R.

APPENDIX 3.—Locations of Chum Salmon in suspected spawning locations in the Koyukuk River Drainage, Alaska 2014.



APPENDIX 4.— Proportion of Chum Salmon with implanted radio transmitters returning to spawning locations in the Koyukuk River Drainage, Alaska, 2015.

	Strata 1 (6/21-6/27)		90% C.I.	
Location	Number Tagged	Proportion	Upper	Lower
Gisasa R.	2	0.069	0.288	0.013
Kateel R.	0	0.000	0.190	0.000
Dulbi R.	0	0.000	0.190	0.000
HusliaR/ Billyhawk Cr.	9	0.310	0.551	0.142
Dakli/Wheeler Cr.	5	0.172	0.411	0.059
Hogatza R.	4	0.138	0.372	0.041
Indian R.	0	0.000	0.190	0.000
Hughes Cr.	0	0.000	0.190	0.000
Alatna R.	0	0.000	0.190	0.000
Henshaw Cr.	7	0.241	0.483	0.098
Koyukuk R.	2	0.069	0.288	0.013
Total	29	1.000		

	Strata 2 (6/28-7/4)		90% C.I.	
Location	Number Tagged	Proportion	Upper	Lower
Gisasa R.	1	0.025	0.185	0.003
Kateel R.	1	0.025	0.185	0.003
Dulbi R.	0	0.000	0.145	0.000
HusliaR/ Billyhawk Cr.	17	0.425	0.625	0.247
Dakli/Wheeler Cr.	4	0.100	0.286	0.030
Hogatza R.	3	0.075	0.255	0.019
Indian R.	1	0.025	0.185	0.003
Hughes Cr.	1	0.025	0.185	0.003
Alatna R.	0	0.000	0.145	0.000
Henshaw Cr.	9	0.225	0.429	0.101
Koyukuk R.	3	0.075	0.255	0.019
Total	40	1.000		

	Strata 3 (7/5-7/11)		90% C.I.	
Location	Number Tagged	Proportion	Upper	Lower
Gisasa R.	6	0.071	0.181	0.026
Kateel R.	2	0.024	0.114	0.005
Dulbi R.	1	0.012	0.096	0.001
HusliaR/ Billyhawk Cr.	21	0.250	0.389	0.149
Dakli/Wheeler Cr.	9	0.107	0.226	0.047
Hogatza R.	11	0.131	0.255	0.062
Indian R.	3	0.036	0.132	0.009
Hughes Cr.	0	0.000	0.075	0.000
Alatna R.	2	0.024	0.114	0.005
Henshaw Cr.	6	0.071	0.181	0.026
Koyukuk R.	23	0.274	0.414	0.168
Total	84	1.000		

APPENDIX 4.—Continued.

	Strata 4 (7/12-7/18)		90% C.I.	
Location	Number Tagged	Proportion	Upper	Lower
Gisasa R.	4	0.121	0.336	0.036
Kateel R.	1	0.030	0.218	0.003
Dulbi R.	1	0.030	0.218	0.003
HusliaR/ Billyhawk Cr.	12	0.364	0.587	0.187
Dakli/Wheeler Cr.	1	0.030	0.218	0.003
Hogatza R.	0	0.000	0.171	0.000
Indian R.	0	0.000	0.171	0.000
Hughes Cr.	0	0.000	0.171	0.000
Alatna R.	0	0.000	0.171	0.000
Henshaw Cr.	2	0.061	0.260	0.012
Koyukuk R.	12	0.364	0.587	0.187
Total	33	1.000		

	Strata 5 (7/19-25)		90% C.I.	
Location	Number Tagged	Proportion	Upper	Lower
Gisasa R.	3	0.188	0.513	0.048
Kateel R.	1	0.063	0.379	0.007
Dulbi R.	0	0.000	0.298	0.000
HusliaR/ Billyhawk Cr.	0	0.000	0.298	0.000
Dakli/Wheeler Cr.	2	0.125	0.449	0.020
Hogatza R.	1	0.063	0.379	0.007
Indian R.	0	0.000	0.298	0.000
Hughes Cr.	0	0.000	0.298	0.000
Alatna R.	0	0.000	0.298	0.000
Henshaw Cr.	3	0.188	0.513	0.048
Koyukuk R.	6	0.375	0.679	0.145
Total	16	1.000		

	Strata 6 (7/26-7/31)		90% C.I.	
Location	Number Tagged	Proportion	Upper	Lower
Gisasa R.	1	0.200	0.722	0.023
Kateel R.	0	0.000	0.576	0.000
Dulbi R.	0	0.000	0.576	0.000
HusliaR/ Billyhawk Cr.	0	0.000	0.576	0.000
Dakli/Wheeler Cr.	1	0.200	0.722	0.023
Hogatza R.	0	0.000	0.576	0.000
Indian R.	0	0.000	0.576	0.000
Hughes Cr.	0	0.000	0.576	0.000
Alatna R.	0	0.000	0.576	0.000
Henshaw Cr.	0	0.000	0.576	0.000
Koyukuk R.	3	0.600	0.919	0.017
Total	5	1.000		